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Completed	Tuesday, 20 October 2026, 9:06 PM
Duration	4 mins 4 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94

b. 94 and 99

c. 90 and 99 ✓

d. 90 and 100

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓

b. Linear search requires the list to be sorted.

c. Linear search divides the list into halves.

d. Linear search is more efficient than binary search on large lists.

Question **3**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search

b. Binary search ✓

c. Hash search

d. Sequential search

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By using a hash function
c. By sorting the list first
d. By comparing each element in the list sequentially ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search continues until the end of the list
c. The search starts over from the beginning
d. The search stops immediately ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Dividing the list in half repeatedly
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **7**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Searching ✓
c. Finding
d. Mining

Question **8**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element ✗
- b. The search stops
- c. The middle element is chosen randomly
- d. The higher middle element is chosen as the middle element

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Hashing
- b. Binary search
- c. Jump search
- d. Linear search ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is very small
- b. When the list contains duplicate elements
- c. When the list is unsorted
- d. When the list is sorted ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- c. Must use a sorted array
- d. There must be a mechanism to access middle element directly

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O👎
b. O(log n) ✓
c. O(1)
d. O(n log n)

Question **13**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓
b. When the list contains duplicate elements
c. When the list is sorted
d. When the search algorithm finishes

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The list is sorted
d. The search continues from the beginning

Question **15**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. At the beginning of the list
c. In the left sublist
d. In the middle of the list